Product Requirements Document - Biometric Attendance Management Platform

Overview

This document specifies the software scope of the Biometric Attendance Management System: the cloud platform, the web dashboard, the integration layer that binds the system to the web software, and the embedded device services such as synchronisation, event ingestion, and real-time messaging.

Executive Statement

The proposed Biometric Attendance Management System will establish a centralized, secure, and scalable platform for managing employee attendance across multiple branches and operational locations. The solution will integrate with the company's existing web software, which will serve as the primary source of truth for employee records, assignments, and employment status.

The project will include the development of supporting terminal services and backend infrastructure required to enable reliable communication between biometric attendance terminals and the central platform. These services will support employee and biometric synchronization, attendance and event ingestion, real-time messaging, offline data synchronization, device monitoring, and over-the-air firmware updates.

The system will be designed with an offline-first architecture, allowing biometric terminals to continue verifying employees and recording attendance even when internet connectivity

is unavailable or unstable. Once connectivity is restored, pending attendance events and synchronization updates will be automatically transmitted to the central system.

Through real-time messaging and centralized device management, the platform will enable biometric terminals to receive updates, synchronize employee and biometric records, report operational status, and receive remote configuration or firmware updates without requiring physical intervention at each location.

The overall solution will provide a reliable and centralized biometric attendance ecosystem that improves attendance visibility, reduces manual intervention, supports multi-location operations, and provides a scalable foundation for future workforce management and biometric technology enhancements.

Objectives and Success Criteria

Objectives

ID	Objective
O-1	Establish one attendance record per employee per location that management can act on without verification
O-2	Remove all manual intermediation between the point of clock-in and the software
O-3	Guarantee that no attendance event is lost when a site is offline or unpowered
O-4	Enforce, in software, that biometric enrolment can only happen at authorised branch devices

Success criteria

ID	Measure	Target
S-1	Attendance event visible in the web software after a clock-in, device online	≤ 10 seconds (p95)
S-2	Attendance events lost or duplicated across a 30-day pilot	Zero
S-3	Employee change made in web software reflected on the target device	≤ 15 minutes on change detection; ≤ 24 hours guaranteed
S-4	Buffered events successfully flushed after a 72-hour outage	100%
S-6	Platform availability, measured monthly	≥ 99.5%

Scope

- Cloud backend: device API, web integration adapter, rules engine, reporting API
- MQTT broker deployment, topic design, per-device credentials and access control
- Web dashboard for administrators, web users and branch supervisors
- Device registry, provisioning and lifecycle management
- Fingerprint template escrow and distribution
- Cloud infrastructure provisioning, domain, TLS, monitoring and six (6) months of hosting
- Documentation: architecture, API reference, integration guide, user guide, runbook

System Architecture

The system uses two channels deliberately. This is a firm architectural decision.

Channel	Protocol	Carries
Durable	HTTPS	Roster sync, fingerprint template upload and download, batched attendance upload, device activation, log upload
Real-time	MQTT	Single attendance event on capture, device presence and heartbeat, roster-changed nudge, remote commands, alert fan-out to dashboard

Functional Requirements

Web software integration

ID	Requirement	Priority
FR-1.1	The adapter shall ingest, per employee: web reference ID, staff number, full name, assigned location reference, employment status, and last-modified timestamp	Must
FR-1.2	The platform shall map every web location reference to exactly one location record, and shall raise an	Must

ID	Requirement	Priority
	unmapped-location alert rather than silently discarding employees	
FR-1.3	On an employee becoming inactive, the platform shall remove them from all device rosters within one sync cycle while retaining their historical attendance records	Must
FR-1.4	The platform shall push attendance events back into the web software with retry, exponential backoff, and a dead-letter queue for events web software rejects	Must
FR-1.5	Where the web software cannot accept programmatic attendance input, the platform shall generate a scheduled export in the web software's accepted import format and deliver it to an agreed endpoint	Must
FR-1.6	Every sync run and a failed sync shall raise an alert to system administrators after two consecutive failures	Must
FR-1.7	A failed sync shall raise an alert to system administrators after two consecutive failures	Should
FR-1.8	Each device shall be provisioned with a unique identity and credential at manufacture, recorded in the device registry	Must

Biometric enrolment

ID	Requirement	Priority
FR-2.1	Enrolment shall require an on-device supervisor authorisation step (supervisor fingerprint or PIN) before an enrolment session may begin	Must

ID	Requirement	Priority
FR-2.2	A device shall not create employee records. Enrolment shall only be possible against an employee already present on the device roster, sourced from the web software	Must
FR-2.3	The device shall capture the fingerprint, generate the template locally, and upload the template to the platform over HTTPS	Must
FR-2.4	The platform shall support enrolment of at least two fingers per employee, to allow for injury and poor-quality prints	Should

Attendance capture and ingestion

ID	Requirement	Priority
FR-3.1	Each attendance event shall record: employee, device, location, direction (in/out), captured timestamp with timezone offset, and match score,	Must
FR-3.2	The ingestion API shall be idempotent. A repeated submission shall return success without creating a duplicate	Must

Offline operation and reconciliation

ID	Requirement	Priority
FR-4.1	The device shall hold a complete local copy of its roster and templates, and shall authenticate and record attendance with no dependency whatsoever on network availability at the moment of capture	Must
FR-4.2	On restoration of connectivity, the device shall automatically upload buffered events in batches over HTTPS, oldest first, without operator action	Must
FR-4.3	When local storage approaches capacity, the device shall raise an alarm and shall preserve oldest events in preference to accepting new ones only after operator-confirmed policy; default policy shall be to continue capturing and alert loudly	Must

Administration

ID	Requirement	Priority
FR-5.1	Dashboard authentication shall use email and password with configurable complexity, session expiry, and account lockout	Must
FR-5.2	Multi-factor authentication shall be available and enforceable for Administrator and Manager roles	Must
FR-5.3	Authorisation shall be role-based and location-scoped, enforced server-side on every request	Must

ID	Requirement	Priority
FR-5.4	Every mutating action by any actor shall be written to an append-only audit log with before/after state	Must
FR-5.5	The audit log shall be searchable and exportable, and shall not be deletable through the application	Must

Technology stack

Layer	Choice
Backend	Node.js 22 LTS, TypeScript
Database	PostgreSQL 16
Cache and Queue	Redis
MQTT broker	Mosquitto with TLS
Object storage	S3-compatible bucket for firmware artefacts and template blobs
Frontend	React, Vite, TailwindCSS
Live updates	WebSocket gateway bridged from the broker
Hosting	Single VPS or small cloud instance, containerised, reverse proxy with TLS
Observability	Structured logging, uptime monitoring, error tracking, database backups

Non-Functional Requirements

Security

NFR-1.1: All HTTPS traffic shall use TLS 1.2 or above with a valid certificate.

NFR-1.2: All MQTT traffic shall use TLS on port 8883 with server certificate validation by the device.

NFR-1.3: Each device shall hold a unique credential.

NFR-1.4: Requests shall carry a nonce or timestamp window to prevent replay attacks involving captured attendance or device payloads.

NFR-1.5: Rate limiting shall be applied to protect the platform against abuse and excessive requests.

Performance

NFR-2.1: On-device fingerprint authentication and attendance decision processing shall be completed in under five seconds and shall remain independent of network availability.

NFR-2.2: When a device is online, an attendance event shall become visible in the centralized dashboard in under five seconds.

NFR-2.3: The attendance management dashboard shall become interactive in under two seconds when accessed through a standard broadband connection.

NFR-2.4: Analytics queries covering up to twelve months of attendance data for a single location shall return results in a short period time.

Data Protection and Retention

NFR-3.1: Fingerprint biometric template data shall be encrypted at rest using encryption keys that are stored and managed separately from the primary database.

NFR-3.2: Biometric templates shall be removed from the central platform and relevant biometric devices within one synchronization cycle after an employee is terminated or removed from the web application.

NFR-3.3: Database backups shall be performed concurrently and retained for thirty days.

Availability and Scalability

NFR-4.1: The centralized platform shall maintain a target availability of at least 99.5% per month, excluding scheduled maintenance periods.

NFR-4.2: In the event of a platform, server, or internet connectivity outage, biometric devices shall continue operating in offline mode, including fingerprint verification and attendance recording, without loss of attendance data.

NFR-4.3: The system architecture shall support growth of more devices and employees without requiring a fundamental architectural redesign.

NFR-4.4: The database schema and system architecture shall support multiple branches, locations, and organizational tenants from the beginning, allowing the platform to accommodate future expansion and multi-organization deployments.